GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 14



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Recap of Previous Lecture



Topic Time & Distance

Topics to be Covered



Topic-1 Problem on Trains

Topic-2 Boats and Streams

Topic-3 Races

[MCQ]

#Q. A man misses a train by 40 minutes if he travels at 30 kmph. If he travels at 40 kmph, then also he misses the train by 10minutes. What is the minimum speed required to catch the train on time?

A 50 km/hr

B 55 km/hr

C 60 km/hr

D 45 km/hr

$\frac{60}{40} = \frac{3}{2} = 1:30$

$\frac{x}{30} - \frac{x}{40} = \frac{1}{2}$

$\Rightarrow \frac{x}{120} = \frac{1}{2}$

$x = 60$

$T = \frac{60}{30} = 2hrs$

Assignment

Exact time $= 11\text{hr}\ 20\text{min} = 11\frac{20}{60} = 11\frac{1}{3} = \frac{34}{3}$

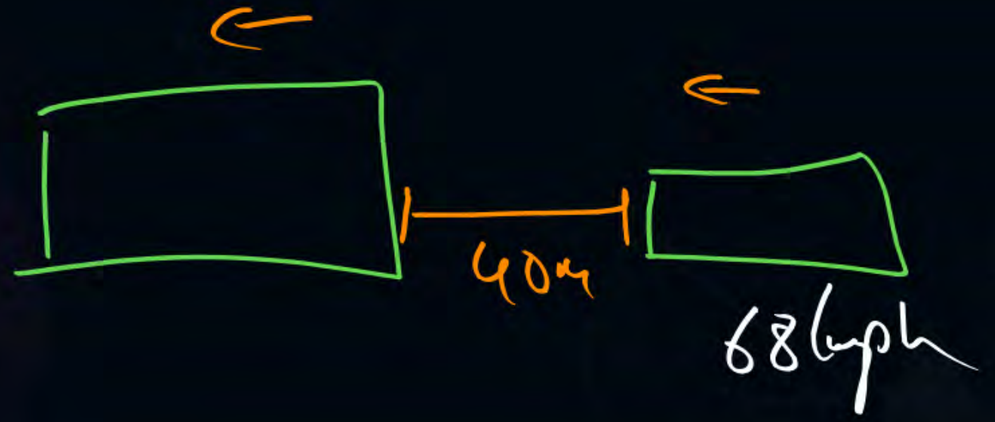
Speed $= \frac{15}{4} \times \frac{3}{4}$

$= 45/16$

[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

Assignment



[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.



$V_D = 100m$

R.S. = ?

$V_T = 10$ sec

$R.S. = \frac{100m}{10sec}$

$= 10 m/sec$

$= 10 \times \frac{18}{5}$ km/hr

$= 36$ km/hr

$(68 - 4)$

$68 - 36 =$ Lorry speed

32 kmph

[MCQ]

#Q. Two boys begin together to write out a booklet containing 817 lines. The first boy starts with first line, writing at the rate of 200 lines an hour and the second boy starts with last line. He writes line 817 and so on backwards proceeding at the rate of 150 lines an hour. At what line will they meet?

A. 469th

B. 467th

C. 465th

D. 463rd

Distance = 817 lines

Relative Speed = 350 lines/hr

Time = $\frac{817}{350}$ hr = ?

Distance = $\frac{817}{350} \times 200 = \frac{3268}{7} = 466.8$

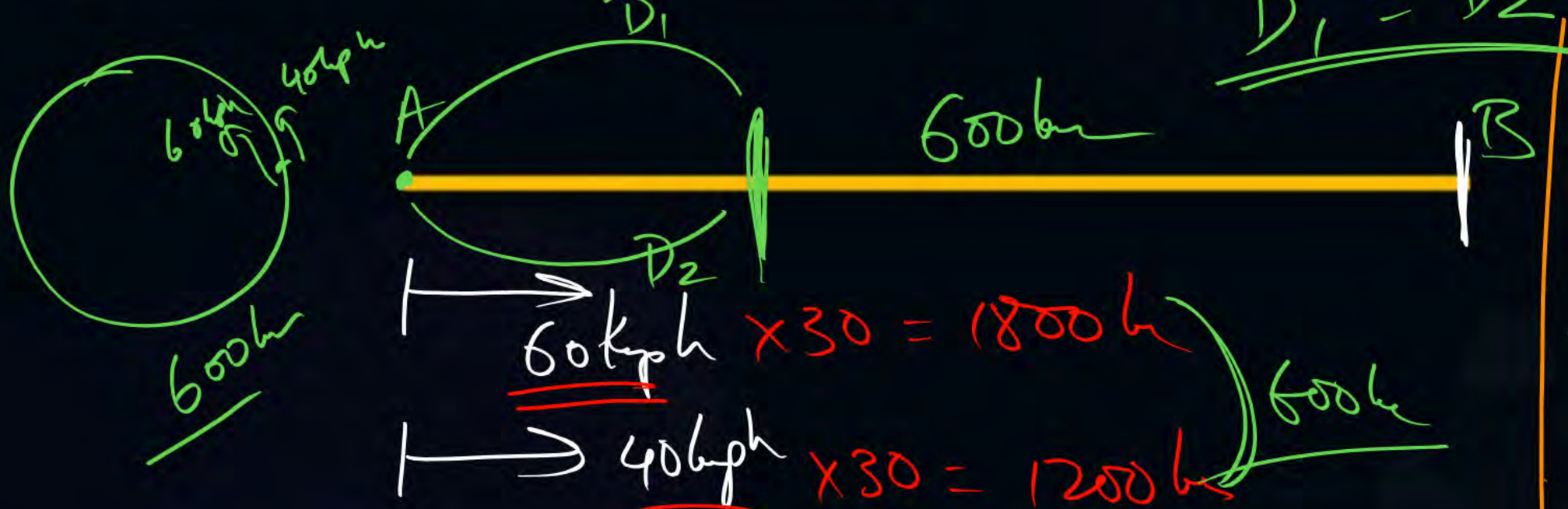
467th

Assignment

Meeting at a point:

$D_1 + D_2 = T.D.$

$D_1 = D_2$



$60 \text{ kmph} \times 30 = 1800 \text{ km}$

$40 \text{ kmph} \times 30 = 1200 \text{ km}$

600km

D = 600km

R.S = 20kmph

$T = \frac{600}{20}$

= 30hrs

Meeting at a point:

Two objects moving in opposite directions towards each other:
Distance covered by 1st object + Distance covered by 2nd object = Total distance





Two objects moving in same directions :
Distance covered by 1st object = Distance covered by 2nd object



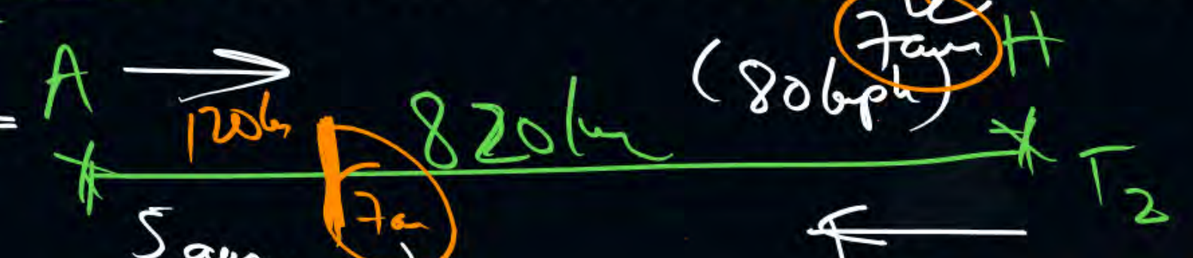


[MCQ]

5am + 7hrs = 12pm

$D_1 + D_2 = T.D.$

#Q. A train starts from Ahmedabad to Howrah at 5am everyday with the speed of 60kmph. Another train starts from Howrah to Ahmedabad at 7am everyday with the speed of 80 kmph. If the distance between Ahmedabad and Howrah is given as 820km, then at what time the two trains meet each other?



$S_{T_1} = 60$kmph

$S_{T_2} = 80$kmph

After 5am → x' hrs

$60x + 80(x-2) = 820$

$\Rightarrow 60x + 80x - 160 = 820$

$140x = 980$

$x = \frac{980}{140} = 7$hrs

D = 700km

RS = 140kmph

$T = \frac{700}{140} = 5$hrs

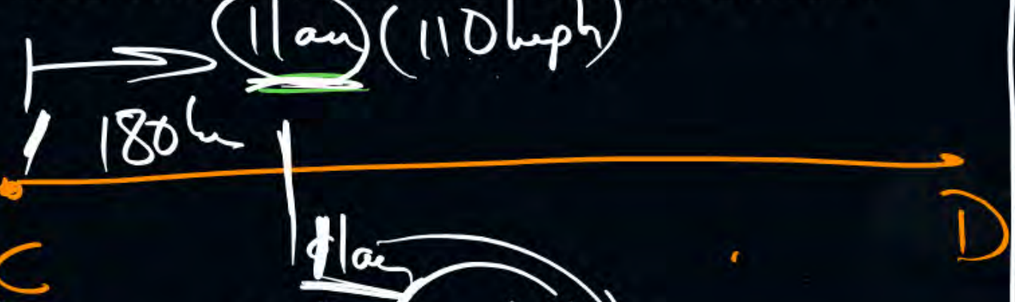
7am + 5hrs = 12pm

[MCQ]

? $D_1 = D_2$

#Q. A train starts from Chennai to Delhi at 11am with the speed of 110kmph. Another train starts from Chennai to Delhi at 9am the same day with the speed of 90 kmph. If the distance between Chennai and Delhi is given as 1920km, then at what time the two trains meet each other?

After 9am → 'x' hrs



$90x = 110(x-2)$

$90x = 110x - 220$

$20x = 220$ hrs

9am + 11hrs = 8pm

D = 180km

R.S = 20kmph

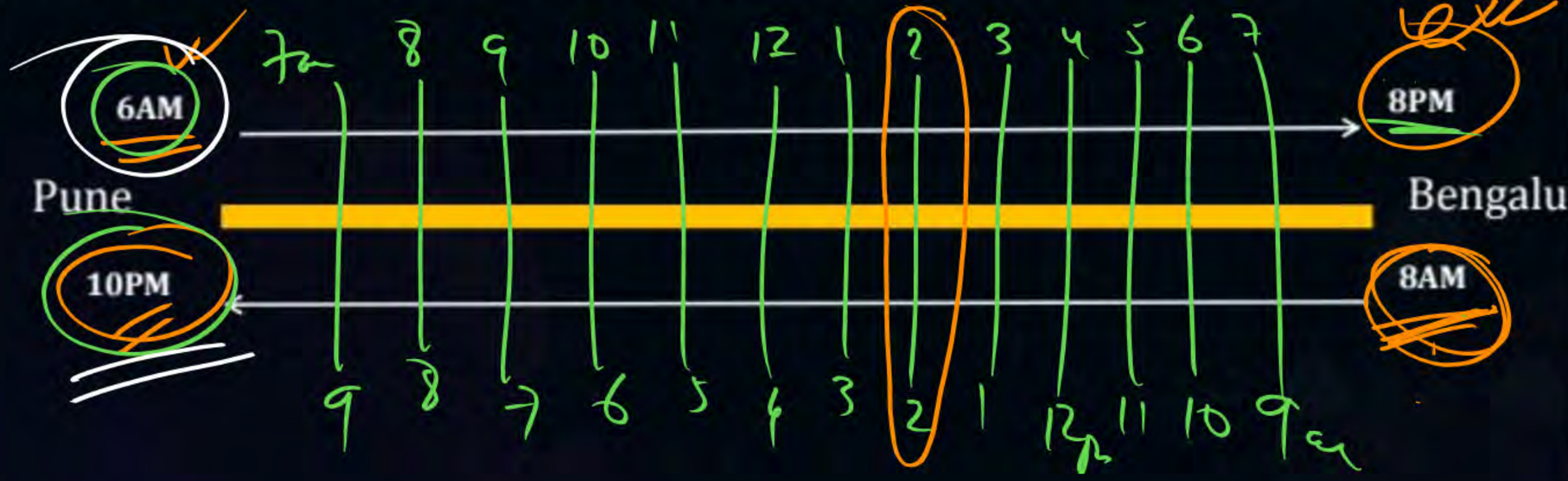
$T = \frac{180}{20} = 9$ hrs

11 + 9 = 20 → 8pm

[MCQ]

2pm

#Q. A train starts from Pune at 6am everyday and reaches Bengaluru 8pm the same day. Another train starts from Bengaluru at 8am everyday and reaches Pune at 10pm the same day. At what time these two trains meet?



6am — 10pm

+8

2pm

8am — 8pm

+6

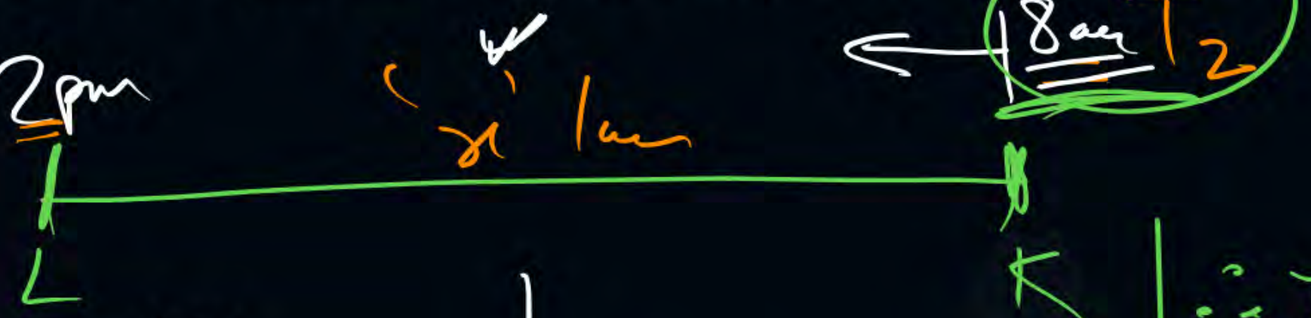
2pm

[MCQ] 5am + 3h = 8:36am $\frac{3}{8} \times \frac{60}{12}$ $D_1 + D_2 = T \cdot D$

#Q. A train starts from Leh at 5am everyday and reaches Kanyakumari at 9am the same day. Another train starts from Kanyakumari at 8am everyday and reaches Leh at 2pm the same day. At what time these two trains meet?

$\delta T_1 = \frac{x}{4}$

$\delta T_2 = \frac{x}{6}$



After (5am + y hrs) T1 5am 9am

$\frac{x}{4} y + \frac{x}{6}(y-3) = x$

$\frac{y}{4} + \frac{y-3}{6} = 1$

$3y + 2y - 6 = 12$

$5y = 18$

$\therefore y = \frac{18}{5}$

$= 3\frac{3}{5}$

8:36 am

After 8 am → 'y' hrs

$$2y + 3y + 9 = 12$$

$$5y = 3$$

$$y = \frac{3}{5} \text{ hrs}$$

$$= \frac{3}{\cancel{5}} \times \cancel{60}^{12} = 36 \text{ min}$$

$$\frac{x}{6} y + \frac{x}{4}(y+3) = x$$

$$\Rightarrow x\left(\frac{y}{6} + \frac{y+3}{4}\right) = x$$

$$\frac{y}{6} + \frac{y+3}{4} = 1$$


5 AM
9 AM
Leh
Kanyakumari
2 PM
8 AM

Trains

D = length of Train



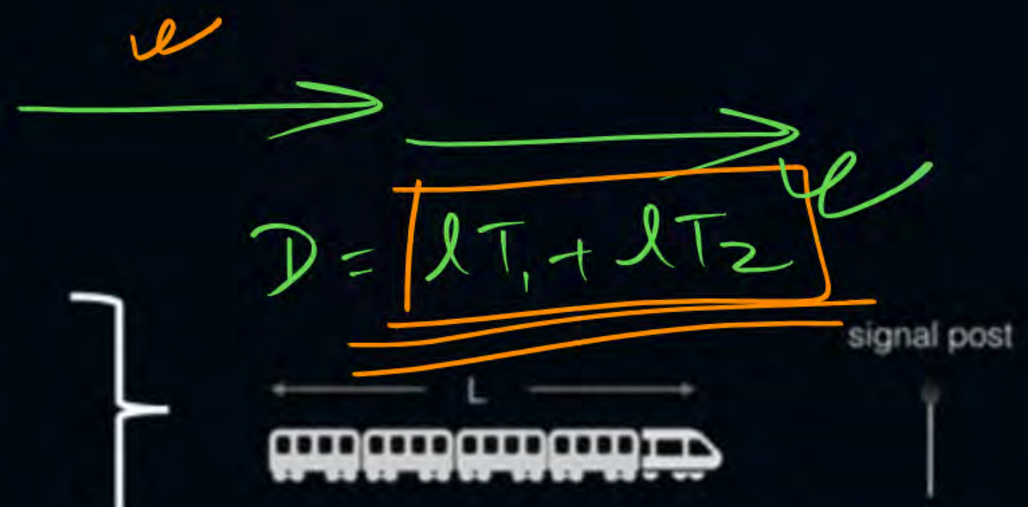


D = l of Train + l of Platform / l Bridge / l of Tunnel



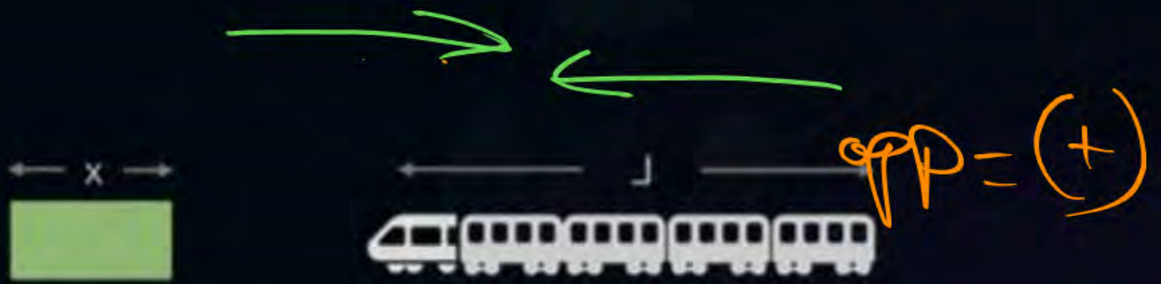
Trains

When a train try to cross a signal pole or standing man completely...
D = length of the train



Same = (-)

When a train try to cross a Platform or Bridge or Tunnel.....
D = length of the train + length of Platform or Bridge or Tunnel



opp = (+)

Trains

When a train try to cross another moving train.....

SAME DIRECTION : Speed (-)
OPPOSITE DIRECTION : Speed (+)

Distance = Length of Train 1 + Length of Train 2



[MCQ]

#Q. If a train travelling at 40 km/hr crosses another train of length 100 meter travelling at 14 km/hr in opposite directions in 30 seconds, then find the length of the train.

Assignment

[MCQ]

#Q. A train running at 52 km/hr takes 36 seconds to pass a platform. Next it takes 24 seconds to cross a man walking at the platform with 10 km/hr in the same direction. Find the length of the platform.

Assignment

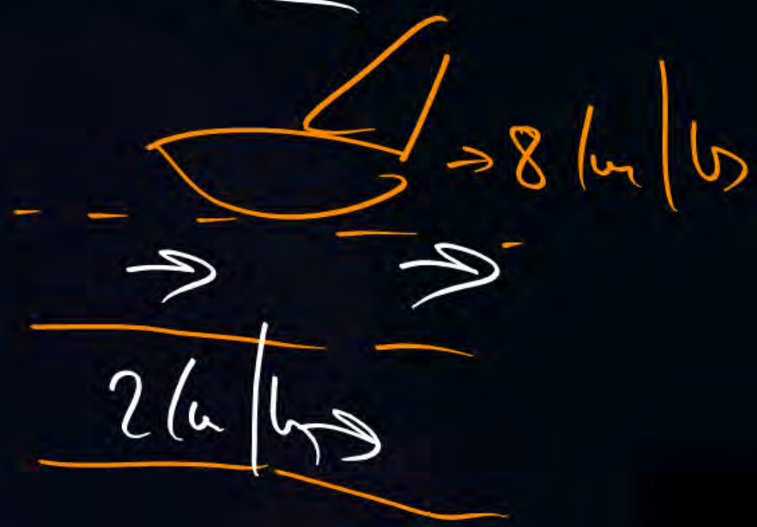
Boats & Streams

(S.B) Speed of Boat = Downstream Speed (DS)

(S.S) Speed of stream = $\frac{\text{Downstream Speed (DS)} - \text{Upstream Speed (US)}}{2}$

Upstream Speed (US)

$= S.B - S.S$



$S.B = \frac{DS + US}{2}$

$S.S = \frac{DS - US}{2}$



[MCQ]

#Q. A man rows his boat downstream @ 18 km/hr & upstream @ 10 km/hr. Find the speed of boat in still water.

Assignment

A 12 km/hr

B 10 km/hr

C 14 km/hr

D 16 km/hr



[MCQ]

#Q. A boat traveled from A to B and back to A from B in 5 hours. If the speed of boat in still water and the speed of stream be 7.5 kmph and 1.5 kmph, then what is the distance between A and B?

Assignment

A 18 km

B 10 km

C 16 km

D 5 km



2 mins Summary

Topic Trains, Boats & Races

THANK - YOU